

A universal limit profile for Steinerberger's amusing sequence

Research note

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Abstract

Consider

$$f_N(x) = \sum_{k=1}^N \frac{|\sin(\pi kx)|}{k}, \quad x \in \mathbb{T} = \mathbb{R}/\mathbb{Z}.$$

Problem 68 in Steinerberger's *Some Open Problems* asks whether the apparently universal shape emerging from f_N can be made precise. We show that after the natural centering by $(2/\pi)H_N$, the sequence converges in $H^s(\mathbb{T})$ for every $s < 1/2$, to an explicit function with exact regularity threshold $1/2$. At every irrational x of finite Diophantine type the convergence is pointwise and the limit is

$$\frac{4}{\pi} \sum_{m=1}^{\infty} \frac{\log(2|\sin(\pi mx)|)}{4m^2 - 1}.$$

For a reduced rational p/q , a different exact logarithmic slope appears:

$$f_N(p/q) = \frac{1}{q} \cot\left(\frac{\pi}{2q}\right) \log N + C_{p,q} + o(1).$$

Thus the universal background and its dense rational valleys are both quantified.

1 Statement of the result

Write $H_N = \sum_{k=1}^N k^{-1}$ and define

$$F_N(x) = f_N(x) - \frac{2}{\pi} H_N.$$

As usual, $\|t\| = \text{dist}(t, \mathbb{Z})$.

Theorem 1 (Universal profile). *There is a real, mean-zero function $F \in L^2(\mathbb{T})$ with the following properties.*

1. For every $s < 1/2$,

$$F_N \longrightarrow F \quad \text{in } H^s(\mathbb{T}).$$

Moreover, $F \notin H^{1/2}(\mathbb{T})$.

2. Suppose x is irrational and there are constants $c > 0$ and $\tau < \infty$ such that

$$\|mx\| \geq cm^{-\tau} \quad (m \geq 1). \tag{1}$$

Then

$$F_N(x) \longrightarrow F(x) = \frac{4}{\pi} \sum_{m=1}^{\infty} \frac{\log(2|\sin(\pi mx)|)}{4m^2 - 1}, \quad (2)$$

where the series converges absolutely. In particular, this applies to almost every x , to every badly approximable irrational, and to every irrational algebraic number.

3. If p/q is reduced, $q \geq 2$, then

$$f_N(p/q) = \mu_q \log N + C_{p,q} + o(1), \quad \mu_q = \frac{1}{q} \cot\left(\frac{\pi}{2q}\right), \quad (3)$$

with

$$C_{p,q} = -\mu_q \log q - \frac{1}{q} \sum_{r=1}^q \left| \sin\left(\frac{\pi pr}{q}\right) \right| \psi\left(\frac{r}{q}\right), \quad (4)$$

where $\psi = \Gamma'/\Gamma$ is the digamma function. In particular,

$$\mu_q = \frac{2}{\pi} - \frac{\pi}{6q^2} + O(q^{-4}). \quad (5)$$

Since $H_N = \log N + \gamma + o(1)$, the first two parts equivalently say that $f_N(x) - (2/\pi) \log N$ approaches $F(x) + 2\gamma/\pi$ at the points covered by (1). At a rational p/q , however,

$$F_N(p/q) = -\left(\frac{2}{\pi} - \mu_q\right) \log N + O(1) \longrightarrow -\infty.$$

The leading depth of the rational valley is $\pi \log N / (6q^2)$ for large q .

2 Sobolev convergence and the exact threshold

The classical absolutely convergent Fourier series

$$|\sin(\pi t)| = \frac{2}{\pi} - \frac{4}{\pi} \sum_{m=1}^{\infty} \frac{\cos(2\pi mt)}{4m^2 - 1} \quad (6)$$

immediately exposes the limit.

Proof of Theorem 1, part 1. Put $g(t) = |\sin(\pi t)| - 2/\pi$. From (6),

$$F_N(x) = \sum_{k=1}^N \frac{g(kx)}{k}.$$

Let $b_{r,N}$ denote the coefficient of $\cos(2\pi rx)$ in this expansion. A frequency r occurs exactly when $r = mk$, and hence

$$b_{r,N} = -\frac{4}{\pi r} \sum_{\substack{m|r \\ r/m \leq N}} \frac{m}{4m^2 - 1}. \quad (7)$$

For each r , this converges monotonically in absolute value to

$$b_r = -\frac{4}{\pi r} \sum_{m|r} \frac{m}{4m^2 - 1}. \quad (8)$$

Since

$$\frac{m}{4m^2 - 1} \leq \frac{1}{3m} \quad \text{and} \quad \sum_{m|r} \frac{1}{m} \leq \sum_{m=1}^r \frac{1}{m} \leq 1 + \log r,$$

we have

$$|b_{r,N}| \leq |b_r| \leq \frac{4(1 + \log r)}{3\pi r}. \quad (9)$$

For every $s < 1/2$, the square of the right-hand side, multiplied by $(1 + r^2)^s$, is summable over r . Dominated convergence of the Fourier coefficients and Plancherel's theorem therefore give $F_N \rightarrow F$ in $H^s(\mathbb{T})$, where F has cosine coefficients b_r .

The divisor $m = 1$ in (8) gives

$$|b_r| \geq \frac{4}{3\pi r}.$$

Consequently the $H^{1/2}$ sum dominates a constant multiple of $\sum_{r \geq 1} r^{-1}$ and diverges. Thus $F \notin H^{1/2}(\mathbb{T})$. The mean of every F_N is zero, so F also has mean zero. $\square$

This also supplies a purely Fourier-series description of the limit:

$$F(x) = -\frac{4}{\pi} \sum_{r=1}^{\infty} \frac{1}{r} \left(\sum_{m|r} \frac{m}{4m^2 - 1} \right) \cos(2\pi r x),$$

with convergence in H^s for $s < 1/2$.

3 Pointwise formula at Diophantine irrationals

We use one standard uniform estimate for logarithmic cosine sums.

Lemma 1. *For $\theta \notin \mathbb{Z}$ and every $N \geq 1$,*

$$\left| \sum_{k=1}^N \frac{\cos(2\pi k \theta)}{k} \right| \leq C \left(1 + \log^+ \frac{1}{\|\theta\|} \right)$$

for an absolute constant C . Moreover,

$$\sum_{k=1}^{\infty} \frac{\cos(2\pi k \theta)}{k} = -\log(2|\sin(\pi \theta)|).$$

Proof. The second identity is the usual Fourier series for the logarithm. Put $\delta = \|\theta\|$. If $N \leq \delta^{-1}$, the trivial estimate by H_N gives the claimed bound. If $N > \delta^{-1}$, compare with the infinite series; summation by parts and the geometric-sum estimate give

$$\left| \sum_{k>N} \frac{e^{2\pi i k \theta}}{k} \right| \leq \frac{C}{N\delta} \leq C.$$

The logarithmic identity completes the proof. $\square$

Proof of Theorem 1, part 2. Using (6), whose series is absolutely convergent, write

$$F_N(x) = -\frac{4}{\pi} \sum_{m=1}^{\infty} \frac{1}{4m^2 - 1} \sum_{k=1}^N \frac{\cos(2\pi mkx)}{k}.$$

Under (1), Lemma 1 bounds the inner sum by $C_{x,\tau}(1 + \log m)$, uniformly in N . This is summable after multiplication by $(4m^2 - 1)^{-1}$. Dominated convergence and the second part of the lemma give exactly (2).

The same estimates show absolute convergence of the series in (2). Almost every x satisfies (1) for any fixed $\tau > 1$ by the Borel–Cantelli lemma. The other stated classes are standard consequences of bounded continued-fraction coefficients and Roth’s theorem, respectively. $\square$

There is also an L^2 interpretation of (2) with no Diophantine hypothesis. The functions $\log(2|\sin(\pi mx)|)$ have an L^2 norm independent of m , and the weights are summable. Thus the logarithmic series converges absolutely in L^2 ; its Fourier coefficients are precisely (8), so it represents the same function F . The hypothesis (1) is used only to justify pointwise convergence of the original sequence at the specified x .

4 The rational asymptotic

Proof of Theorem 1, part 3. Set

$$a_r = \left| \sin\left(\frac{\pi pr}{q}\right) \right|, \quad 1 \leq r \leq q.$$

The sequence $|\sin(\pi kp/q)|$ is q -periodic. Since multiplication by p permutes the nonzero residues modulo q ,

$$\begin{aligned} \frac{1}{q} \sum_{r=1}^q a_r &= \frac{1}{q} \sum_{r=1}^{q-1} \sin\left(\frac{\pi r}{q}\right) \\ &= \frac{1}{q} \cot\left(\frac{\pi}{2q}\right) = \mu_q. \end{aligned}$$

For $N = qM$, grouping the harmonic sum by residue class and using the digamma identity gives

$$\begin{aligned} f_{qM}(p/q) &= \frac{1}{q} \sum_{r=1}^q a_r \left[\psi\left(M + \frac{r}{q}\right) - \psi\left(\frac{r}{q}\right) \right] \\ &= \mu_q \log M - \frac{1}{q} \sum_{r=1}^q a_r \psi\left(\frac{r}{q}\right) + o(1). \end{aligned}$$

Replacing $\log M$ by $\log N - \log q$ proves (3)–(4); the bounded incomplete final block shows that the same asymptotic holds for arbitrary N . Finally, the Taylor series for $\cot z$ at zero yields (5). $\square$

Interpretation and novelty caveat

The limit F is a concrete version of the universal object suggested by the plots in Problem 68. It has critical Sobolev regularity $1/2$ and logarithmic singularities at every rational: in (2), the term $m = q$ already diverges when $x = p/q$. The finite functions F_N regularize these singularities, producing the increasingly fine rational cusp pattern.

Steinerberger’s original paper proves that every rational p/q becomes a strict local minimum once $N \geq q^2$ and obtains the asymptotically sharp threshold $N \geq (1 + o(1))q^2/\pi$; it does not study the renormalized limit. The January 2025 problem list still asks for a precise description of the universal function. A targeted search through July 20, 2026 located only the original result and works citing its local-minimum phenomenon, not the limit theorem above. Because the ingredients are classical, this note does not claim an exhaustive novelty search.

References

- [1] S. Steinerberger, *An amusing sequence of functions*, Mathematics Magazine **91** (2018), 262–266; [arXiv:1610.04090](#).
- [2] S. Steinerberger, *Some Open Problems*, problem 68, version dated January 6, 2025, [Xena Project problem-list](#) page.